

LUCKY BADSRA

India | +91 7568929460 | badsrалucky@gmail.com

LinkedIn: [lucky-badsra](#) | GitHub: [badsrалucky](#)

PROFESSIONAL SUMMARY

M.E. Computer Science student building applied AI and ML projects end-to-end, from a RAG-based document Q&A system with FastAPI and FAISS to a ResNet18 pneumonia classifier deployed on Streamlit. Comfortable across the full stack (MongoDB, Express, React, Node.js) and quick to pick up new tools when a project needs them.

EDUCATION

M.E. in Computer Science Engineering 2025 – 2027 (Expected)

Thapar Institute of Engineering & Technology | CGPA: 8.09

B.Tech in Computer Science Engineering 2020 – 2024

SRM College, Kurukshetra University | CGPA: 8.54

TECHNICAL SKILLS

- **Generative AI & LLMs:** Retrieval-Augmented Generation (RAG), Llama 3.2, Ollama, LoRA, Sentence Transformers, FAISS, Prompt Engineering, Agentic AI workflows
- **AI & Machine Learning:** PyTorch, Scikit-learn, Pandas, NumPy, CatBoost, ResNet18 (CNN), Model Evaluation
- **Programming Languages:** Python, C++, JavaScript, HTML, CSS
- **Web & Automation:** FastAPI, MongoDB, Express, React, Node.js, REST APIs
- **Developer Tools:** Git, GitHub, VS Code, Postman

TECHNICAL PROJECTS

AI Research & Localization Studio | *Python, FastAPI, FAISS, Ollama, Llama 3.2* May 2026 – Present

- Built an end-to-end RAG application that lets users query uploaded documents in natural language, reflecting the AI-assisted knowledge management and engineering copilot tools this role focuses on.
- Engineered document ingestion and semantic search using FastAPI, Sentence Transformers, and a FAISS vector database, reducing manual document lookup effort.
- Integrated Llama 3.2 via Ollama to generate grounded, context-aware responses; followed a modular backend architecture for rapid iteration and future tool integration.

Explainable Pneumonia Detection from Chest X-rays | *ResNet18, PyTorch, Streamlit, Python* Jan 2026

- Developed a deep learning app to classify chest X-rays using ResNet18 transfer learning, trained with image augmentation, CrossEntropyLoss, and the Adam optimizer.
- Built a Streamlit interface with confidence scoring, downloadable reports, prediction history, and Grad-CAM visual explanations for interpretability.

Indoor Plant Health & Growth Analysis | *Python, CatBoost, Scikit-learn* Jun 2025

- Built a CatBoost-based predictive analytics pipeline to classify plant health status (healthy/stressed) from environmental and growth parameters such as soil moisture, light exposure, temperature, humidity, and nutrient levels.
- Performed data cleaning, feature engineering, and exploratory analysis on the growth dataset to surface relationships between environmental conditions and plant outcomes.
- Ran feature-importance analysis to identify the environmental factors most responsible for plant growth and health outcomes, visualized to support agronomy-style decision-making.
- Validated model performance using Stratified 5-Fold Cross-Validation to ensure reliable, generalizable predictions across varied conditions.

Doctor Appointment Booking Application | *MongoDB, Express, React, Node.js* 2023

- Built a full-stack platform with separate patient and doctor/admin dashboards for browsing doctors by specialty, checking real-time slot availability, and booking or cancelling appointments.
- Implemented secure, token-based user authentication with role-based access control for patients, doctors, and admins, backed by RESTful APIs for appointment, profile, and schedule management.
- Designed a responsive React UI with appointment history, profile editing, and an admin panel for managing doctor listings and confirming bookings.

CERTIFICATIONS

- **C++ Programming with OOPs** – Internshala Trainings, 2022; hands-on training in OOP principles including classes, inheritance, polymorphism, and encapsulation.

LEADERSHIP & ADDITIONAL ACHIEVEMENTS

- **Service Selection Board (SSB):** Vetted for Officer-Like Qualities (OLQs) including leadership, team cohesion, and adaptability for the Indian Army and Indian Navy, reaching the final interview stage among the top percentile of candidates.
- **Indian Coast Guard:** Successfully qualified the written stage for Assistant Commandant (Officer) selection.